R13

Code No: 115EH

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August/September - 2024
OPERATING SYSTEMS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	What is a Kernel?	[2]
b)	What is Dual Mode Operation and what is the need of it?	[3]
c)	Differentiate Process and Thread.	[2]
d)	Brief Thread Scheduling.	[3]
e)	Define Swapping.	[2]
f)	What are internal and external fragmentations?	[3]
g)	What are the allocation methods of a disk space?	[2]
h)	What are the objectives of a file management system?	[3]
i)	Give necessary conditions for Deadlock formation.	[2]
j)	Write about Access matrix.	[3]

PART - B

(50 Marks)

2. List and explain the different types of operating systems. [10]

OR

3.a) State the functions of an OS.
b) What is meant by spooling? [5+5]

4. What is the average waiting time and average turn-around times of all processes for FCFS, SJF algorithm? [10]

Processes	Burst Time	Priority
P1	2	2
P2	1	1
P3	8	4
P4	4	2
P5	5	3

OR

5. Explain about Multiple-Processor Scheduling and Real Time Scheduling. [10]

6.a) Write about LRU Page Replacement algorithm.
b) Differentiate Paging and Segmentation. Give their merits and demerits. [5+5]

OR

7. Discuss in detail about Thrashing and brief the methods to avoid Thrashing. [10]

8. Explain the following with respect to file.
   a) Concept of file
   b) Access methods
   c) File sharing. [3+3+4]

OR

9. Give overview of Mass Storage Structure in detail. [10]

10. How Deadlock can be Detected and Recovered? Give suitable example. [10]

OR

11.a) Write short notes on Capability-Based Systems.
   b) Write short notes on access lists. [5+5]

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